Labour Demand 1: Competitive Labour Markets

EC306- Wages and Employment

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Goals of This Section

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Today we will:

- Understand how firms decide how much labour to employ
- Compare short-run vs. long-run labour demand decisions
- Explain why labour demand is downward sloping
- Learn factors that affect the elasticity of labour demand
- Understand the impact of free trade and international competition on the Canadian labour market

Background

Background

Labour demand: the number of workers firms want to hire at a given wage rate

> **Derived Demand**
>
> Labour demand is linked to demand for goods/services produced — it is an input into that production

Several factors influence labour demand:

- Firm's objectives and constraints
- Demand conditions in product markets
- State of technology
- Time horizon
  - Short run: one or more inputs fixed (usually capital)
  - Long run: all inputs variable

Structure of Product Markets

- Labour demand is driven by the firm's output (labour is an input)
- Structure of the product market influences demand for labour
  - Ranges from perfect competition (many sellers) to monopoly (one seller)
- Structure of the labour market influences supply the firm faces
  - Ranges from perfect competition (many workers) to monopsony (one buyer)

> **Note**
>
> In this section we examine perfect competition in both product and labour markets

Demand in the Short Run

Model

As labour demand is derived from production, the model starts with a production function:

$$Q = f(K, N)$$

Where:

- Q is quantity of output
- K is capital (fixed in short run at K_0)
- N is labour (variable in short run)

Firm's objective: maximize profit

In the short run this means choosing N given fixed K_0

Model

To see this mathematically:

Output prices are given (perfect competition)

Profit:

$$\pi(N) = pQ - wN - rK_0$$

$$= pF(K, N) - wN - rK_0$$

Differentiate $\pi(N)$ w.r.t. N:

$$\frac{d\pi}{dN} = p\frac{\partial F}{\partial N}(K_0, N) - w = pMP_N - w$$

Model

Optimality Condition

Optimal N^* satisfies: $pMP_N = w$

Where:

- MP_N is the marginal product of labour
- pMP_N is the marginal revenue product of labour (MRP)
  - When firm is in perfectly competitive product market and $MR = p$ it is called value of marginal product (VMP)
- w is the wage rate (marginal cost of labour, MC_N)

Optimality requires that the value of the last unit of labour hired equals its cost

Model

Other key metrics:

- Total Revenue Product (TRP): Total revenue associated with labour employed

$$TRP = p \times Q$$

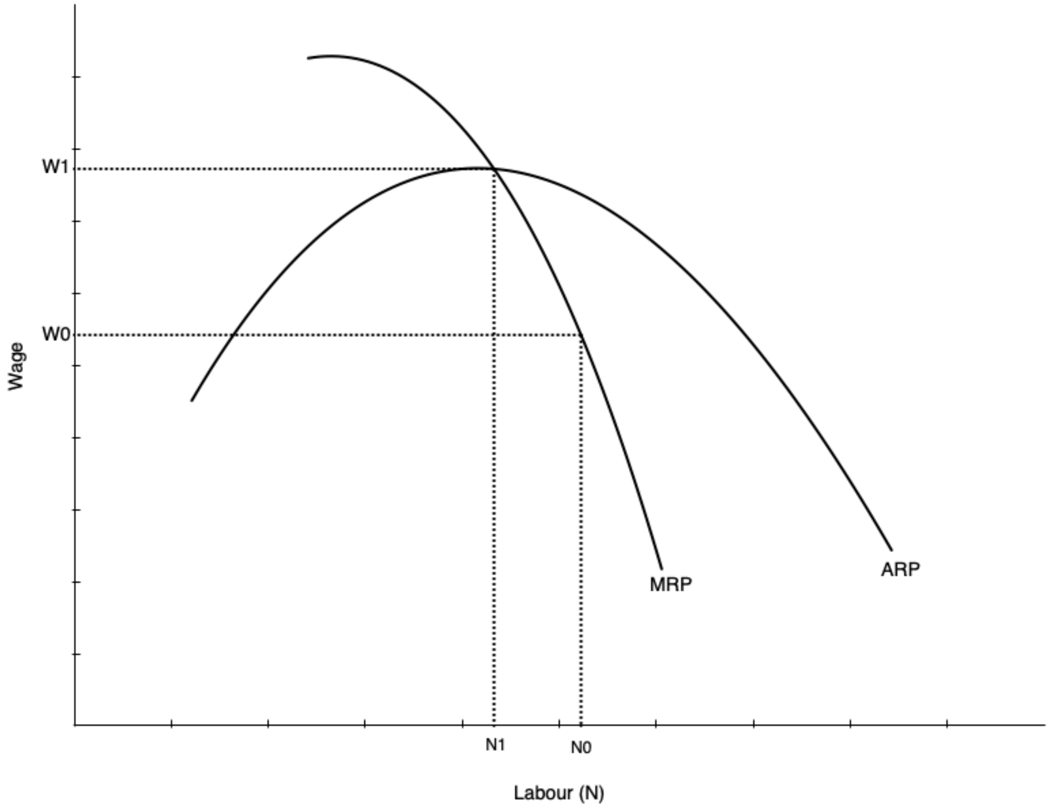
- Average Revenue Product (ARP): Average revenue per unit of labour employed

$$ARP = \frac{TRP}{N} = \frac{p \times Q}{N}$$

The firm's optimal choice of labour is where $MRP_N = w$

- This defines its labour demand curve in the short run
- As w varies, the firm adjusts N to maintain equality

Labour Demand Curve in Short Run



The short run labour demand curve is the MRP curve below ARP

- Shows amount of labour demanded at every wage
- Slopes down because of diminishing marginal product of labour
  - As more labour is employed, extra output produced falls
  - Not because of inferior labour, but because more labour uses fixed capital

As wage falls, firm is willing to use more labour

Demand Curve and Competition in Product Market

- Labour demand depends on demand for the product being produced
- Also depends on structure of the product market
- Firm always maximizes profit by choosing labour where $MRP_N = w$

But marginal revenue differs across market structures:

- In perfect competition: $MR = p$, and MR does not fall with more N
- In monopoly: $MR < p$, and MR falls as N increases

> ⚠ **Warning**
>
> With less competition in product market, labour demand is lower and falls faster with more N

Demand in the Long Run

Introduction

> **Long Run Differs from Short Run**
>
> All inputs are variable — firms can adjust capital as well as labour

This makes the decision more complicated:

- Must consider substitution between labour and capital
- Must consider economies of scale

Approach to find optimum occurs in two steps:

1. Minimize cost holding output constant
2. Choose output to maximize profit given cost function

Cost Minimization

First step: minimize cost of producing given output Q_0

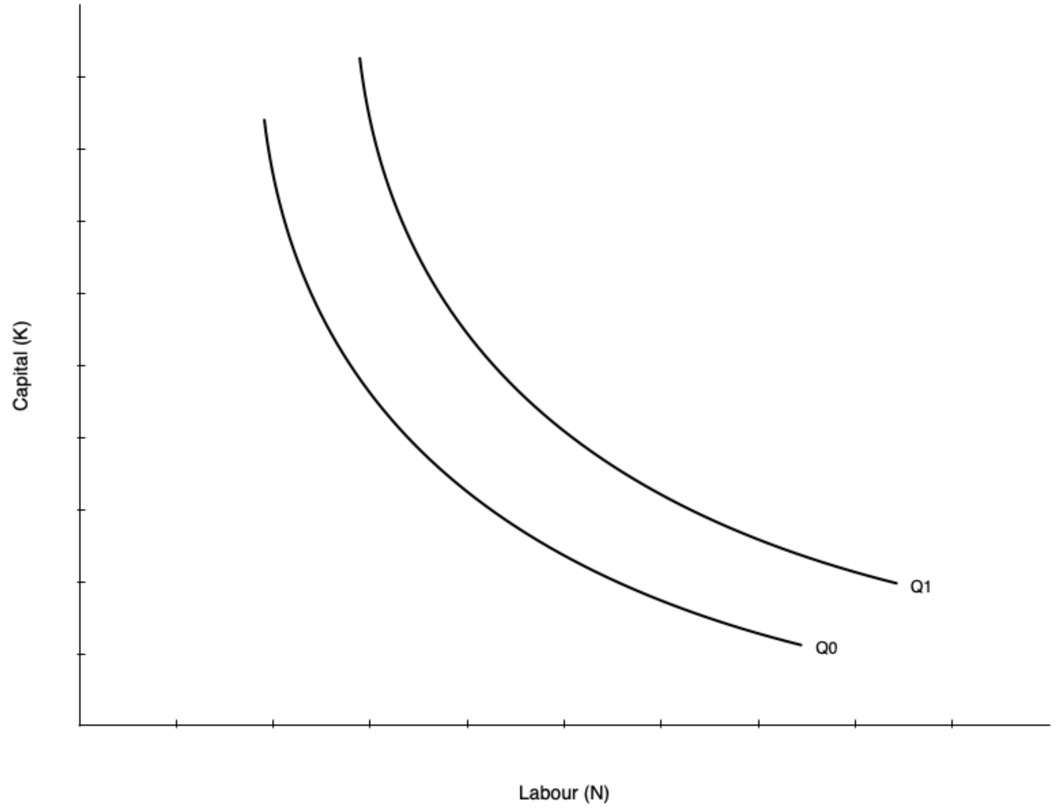
Occurs when an isoquant is tangent to an isocost line:

- Isoquant: shows combinations of K and N that produce Q_0
- Isocost line: shows combinations of K and N that cost the same amount
- Analogous to utility maximization with indifference curves and budget constraints

Assume cost of capital is r and cost of labour is w

- Total cost: $C = rK + wN$
- Production function: $Q = F(K, N)$
- Minimize C subject to $Q = Q_0$

Cost Minimization



Graph shows two isoquants

- Combinations of capital and labour that produce the same quantity
- Shape given by the production function
  - Shown as convex, but could be different shapes
  - Linear isoquant → perfect substitutes
  - L-shaped isoquant → perfect complements

Slope of isoquant is the MRTS

- Marginal rate of technical substitution
- Rate at which you can substitute labour for capital holding output constant

Cost Minimization

The slope of the isoquant is given by the MRTS:

$$MRTS = \frac{MP_N}{MP_K}$$

Tells you amount of capital you can give up for one more unit of labour holding output constant

Diminishing MRTS

Isoquants typically display diminishing MRTS — when a lot of capital is used, you can give up a lot for one more unit of labour; when a lot of labour is used, you can give up only a little capital

Cost Minimization

The isocost line gives combinations of K and N that cost the same amount

Total cost:

$$C = rK + wN$$

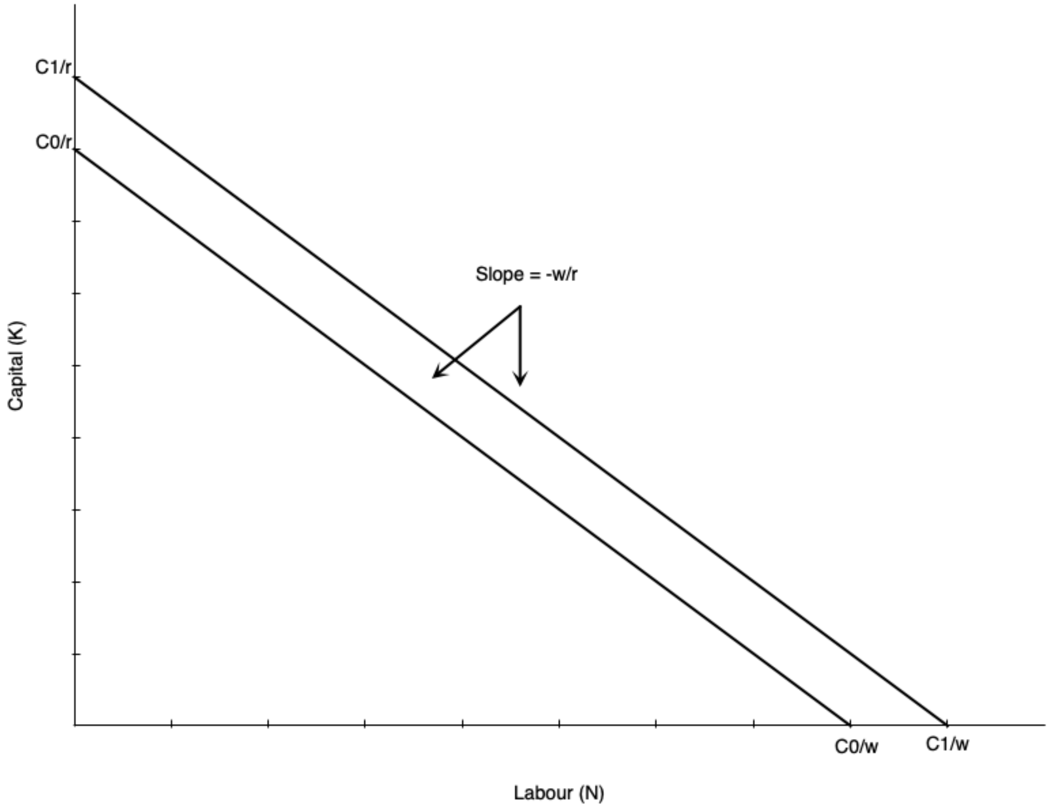
Rearranging gives the isocost line equation:

$$K = \frac{C}{r} - \frac{w}{r}N$$

Slope of isocost line: ratio of input prices

- As w increases, isocost line becomes steeper
- As r increases, isocost line becomes flatter

Cost Minimization

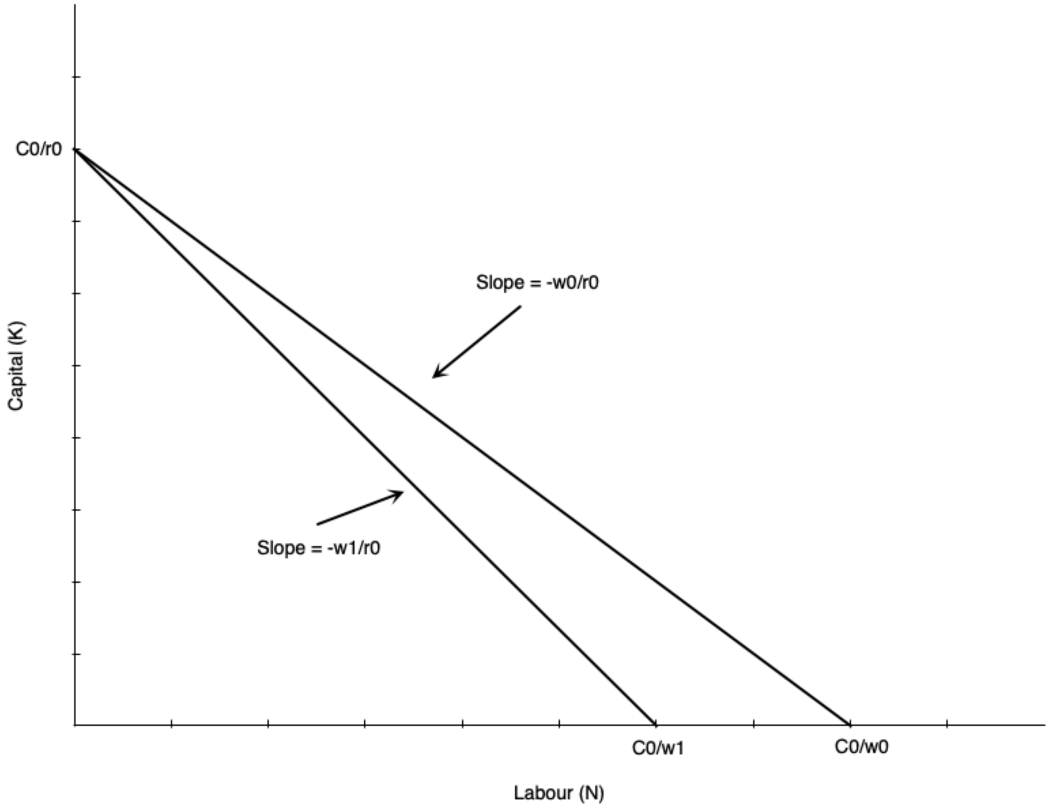


Example isocost:

- Slope is ratio of input prices $-w/r$
- If firm uses only labour: C/w
- If firm uses only capital: C/r
- As labour falls by one unit, capital must rise by w/r units to keep cost constant

Higher cost line shifts isocost outward (when C_0 increases to C_1)

Cost Minimization



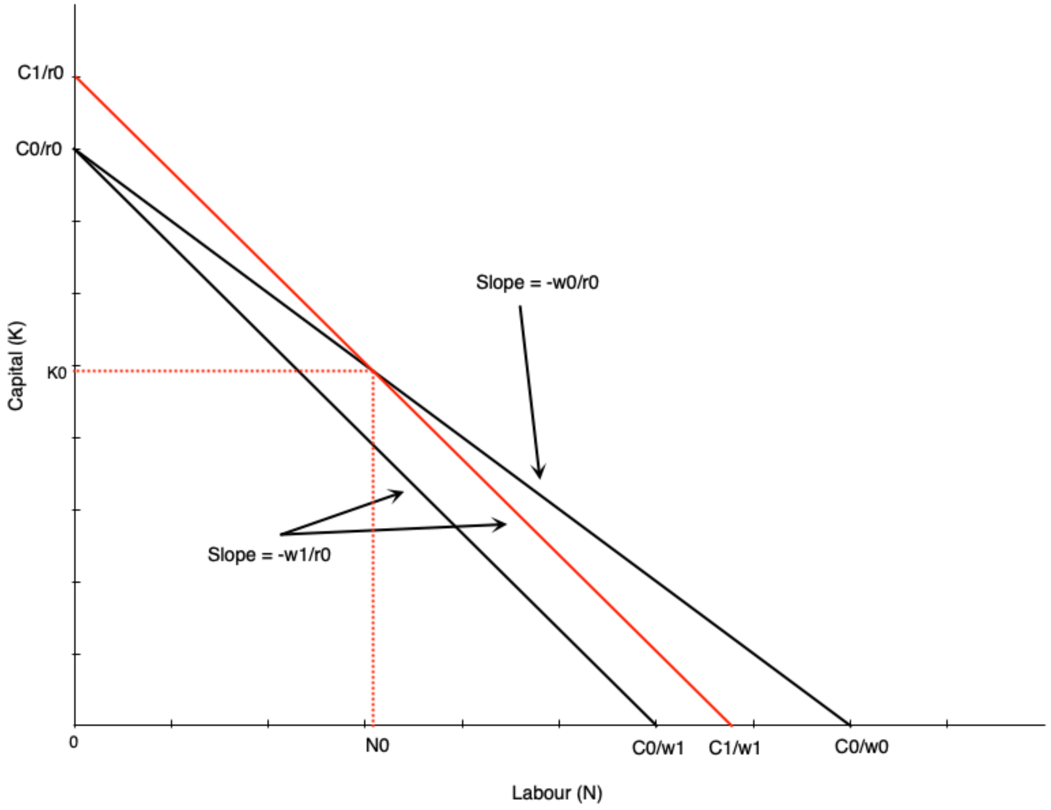
Suppose wage rises from w_0 to w_1

Holding cost constant at C_0:

- Isocost line pivots inward
- Steeper slope as w/r increases
- If firm uses only capital, vertical intercept stays at C_0/r
- If firm uses only labour, horizontal intercept falls to C_0/w_1

All combinations of new isocost lie below old isocost

Cost Minimization



To use any of the old combinations of K and N firm must increase cost

- Shifts isocost outward to C_1
- Magnitude of shift depends on which combination is chosen

Example depicted shows cost curve shifting out to use combination (N_0, K_0) at new prices

Cost Minimization

Cost Minimizing Bundle

Where isoquant is tangent to isocost line

In mathematical terms:

$$\frac{MP_N}{MP_K} = \frac{w}{r}$$

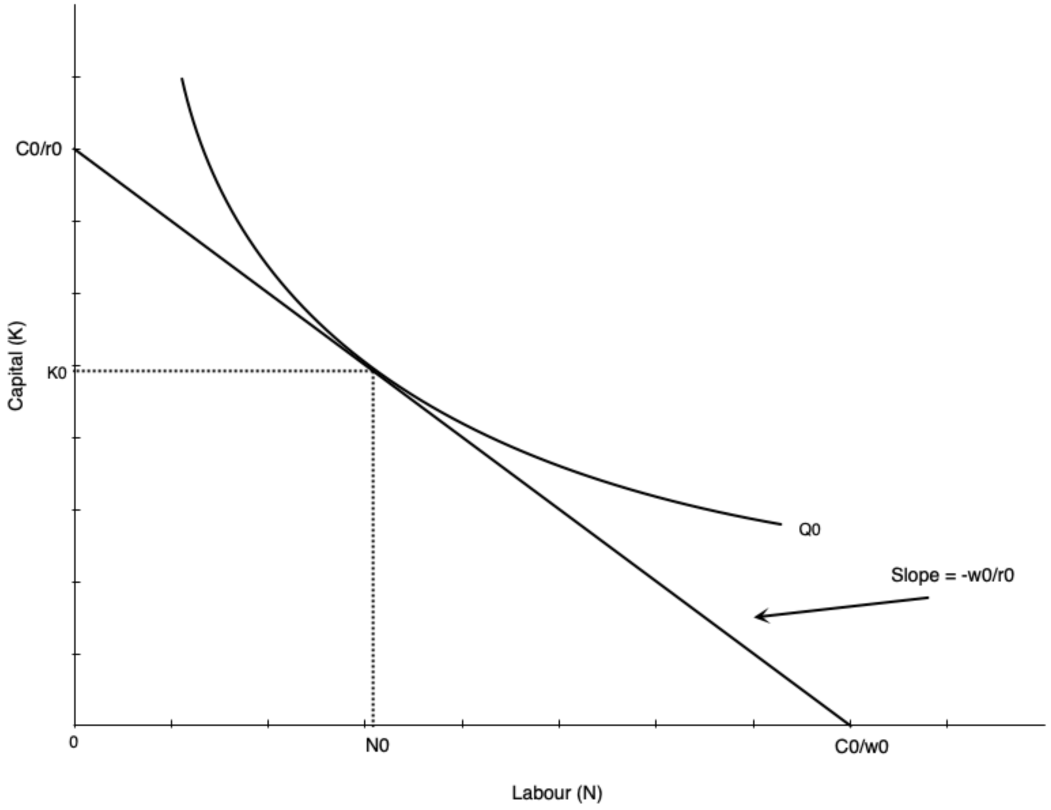
Says that MRTS equals the ratio of input prices

Alternatively:

$$\frac{MP_N}{w} = \frac{MP_K}{r}$$

The last dollar spent on each input must yield the same marginal product

Cost Minimization



Cost minimization depicted

- Hold output constant at Q_0
- Find optimal combination of K and N
- Happens at tangency point between isoquant and isocost line

Cost Minimization

> **Two-Step Process**
>
> We have only done cost minimization here — lowest cost given output level Q_0. Profit maximization involves choosing output level as well.

- Second step: choose Q to maximize profit given cost function
- Second step is not depicted here

If wage rate goes up, firm re-optimizes:

- Firm marginal and average cost curves shift up
- Each firm reduces output
- Market price rises
- Isoquants at profit max shift inward

Cost Minimization – Math

Suppose production exhibits decreasing returns to scale:

$$y = F(K,N) = K^{\alpha}N^{\beta}, \quad \alpha > 0,\ \beta > 0,\ \alpha + \beta < 1$$

Input prices: r (capital), w (labour)

Cost-minimization problem:

$$\min_{K,N}\ C = rK + wN \quad \text{s.t.} \quad K^{\alpha}N^{\beta} \geq y$$

Lagrangian:

$$\mathcal{L} = rK + wN + \lambda\left(y - K^{\alpha}N^{\beta}\right)$$

Cost Minimization – Math

First-order conditions:

$$\frac{\partial \square}{\partial K} = r - \lambda \alpha K^{\alpha-1} N^{\beta} = 0$$

$$\frac{\partial \square}{\partial N} = w - \lambda \beta K^{\alpha} N^{\beta-1} = 0$$

$$y = K^{\alpha} N^{\beta}$$

Divide the first two conditions to eliminate λ:

$$\frac{\beta}{\alpha} \frac{K}{N} = \frac{w}{r}$$

This is the familiar result: $\frac{MP_N}{MP_K} = \frac{w}{r}$

Cost Minimization – Math

From production $y = K^{\alpha} N^{\beta}$:

$$N = \left(\frac{y}{K^{\alpha}} \right)^{\frac{1}{\beta}}$$

Then:

$$\frac{K}{N} = K \left(\frac{K^{\alpha}}{y} \right)^{1/\beta} = K^{\frac{\alpha+\beta}{\beta}} y^{-1/\beta}$$

Cost Minimization – Math

Plug into $\frac{\beta}{\alpha}\frac{K}{N} = \frac{w}{r}$:

$$\frac{\beta}{\alpha} K^{\frac{\alpha+\beta}{\beta}} y^{-1/\beta} = \frac{w}{r}$$

$$K^{\frac{\alpha+\beta}{\beta}} = \frac{\alpha}{\beta}\frac{w}{r} y^{1/\beta}$$

Solving for optimal K:

$$K^*(y; w, r) = y^{\frac{1}{\alpha+\beta}} \left(\frac{\alpha w}{\beta r}\right)^{\frac{\beta}{\alpha+\beta}}$$

Cost Minimization – Math

Use $\dfrac{K}{N} = \dfrac{\alpha w}{\beta r}$ to get N^*:

$$N^*(y; w, r) = \frac{\beta r}{\alpha w} K^* = y^{\frac{1}{\alpha+\beta}} \left(\frac{\alpha\, w}{\beta\, r} \right)^{-\frac{\alpha}{\alpha+\beta}}$$

The cost function:

$$C(w, r, y) = rK^*(y; w, r) + wN^*(y; w, r) = \phi(w, r)\, y^{\frac{1}{\alpha+\beta}}$$

Cost Minimization – Math

Where:

$$\phi(w,r) = r\left(\frac{\alpha w}{\beta r}\right)^{\frac{\beta}{\alpha+\beta}} + w\left(\frac{\alpha w}{\beta r}\right)^{-\frac{\alpha}{\alpha+\beta}}$$

Cost Minimization – Math

Stage 2: choosing output level to maximize profit

Suppose firm is a price taker in product market with price p

Profit:

$$\pi(y) = py - C(y; w, r) = py - \phi(w, r)\, y^{\frac{1}{\alpha+\beta}}$$

Optimize profit with respect to y:

$$\frac{d\pi}{dy} = p - \phi(w, r) \cdot \frac{1}{\alpha + \beta} \cdot y^{\frac{1}{\alpha+\beta} - 1}$$

Optimal output level (set $\frac{d\pi}{dy} = 0$):

$$y^*(w, r, p) = \left(\frac{(\alpha + \beta)\, p}{\phi(w, r)} \right)^{\frac{\alpha+\beta}{1-(\alpha+\beta)}}$$

Cost Minimization – Math

You can then plug in to get optimal inputs:

$$K^*(w,r,p) = \left(\frac{(\alpha+\beta)\,p}{\phi(w,r)}\right)^{\frac{1}{1-(\alpha+\beta)}} \left(\frac{\alpha\, w}{\beta\, r}\right)^{\frac{\beta}{\alpha+\beta}}$$

$$N^*(w,r,p) = \left(\frac{(\alpha+\beta)\,p}{\phi(w,r)}\right)^{\frac{1}{1-(\alpha+\beta)}} \left(\frac{\alpha\, w}{\beta\, r}\right)^{-\frac{\alpha}{\alpha+\beta}}$$

Cost Minimization – Math

⚠ Don’t Get Bogged Down

I will not ask you to recreate this derivation on exams

Key points:

- First minimize cost → optimal inputs as function of output and input prices
  - As you vary any of those three things, the inputs change
- Given cost function, choose output to maximize profit
  - Output depends on product price and input prices
  - Changes in input prices change both optimal inputs and output level

Deriving Labour Demand in Long Run

Deriving Labour Demand

The labour demand curve plots combinations of w and N such that firm is maximizing profit

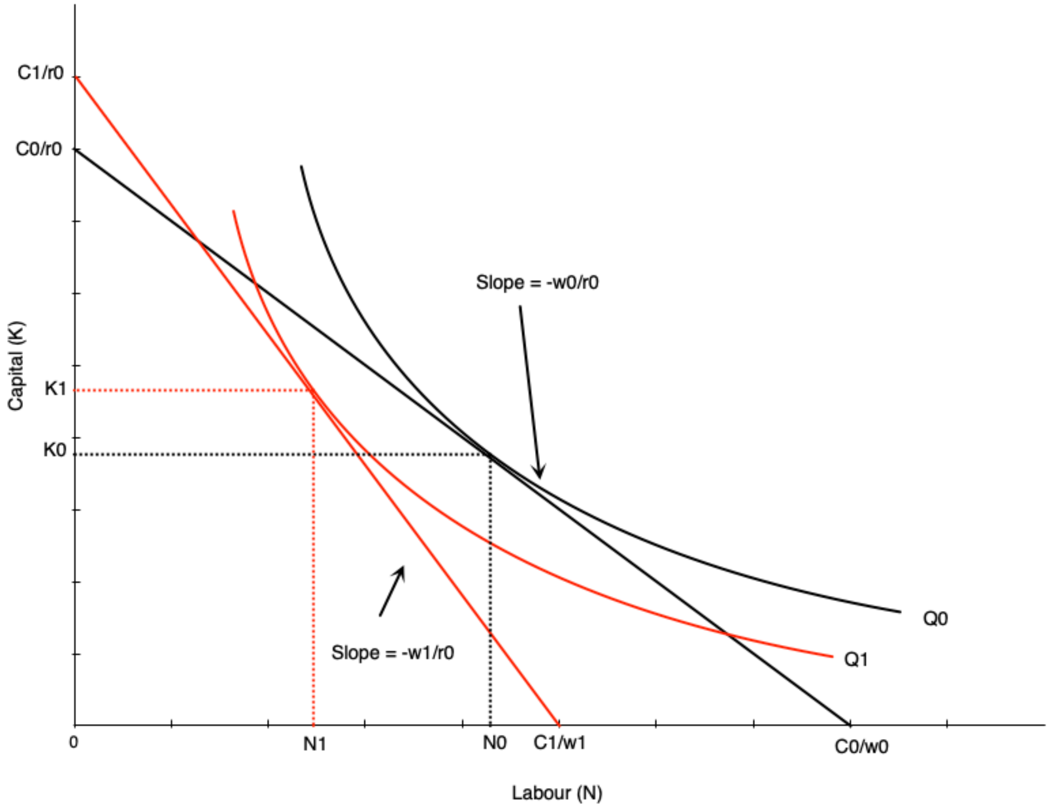
To derive it, consider changing wage from w_0 to w_1:

- Isocosts get steeper as w/r increases
- Firm finds cost minimizing combination of K and N at new wage for each output level
- Because cost increases, firm reduces output (optimal isoquant shifts left)

At new optimum:

- Lower quantity of labour demanded
- Higher wage

Deriving Labour Demand

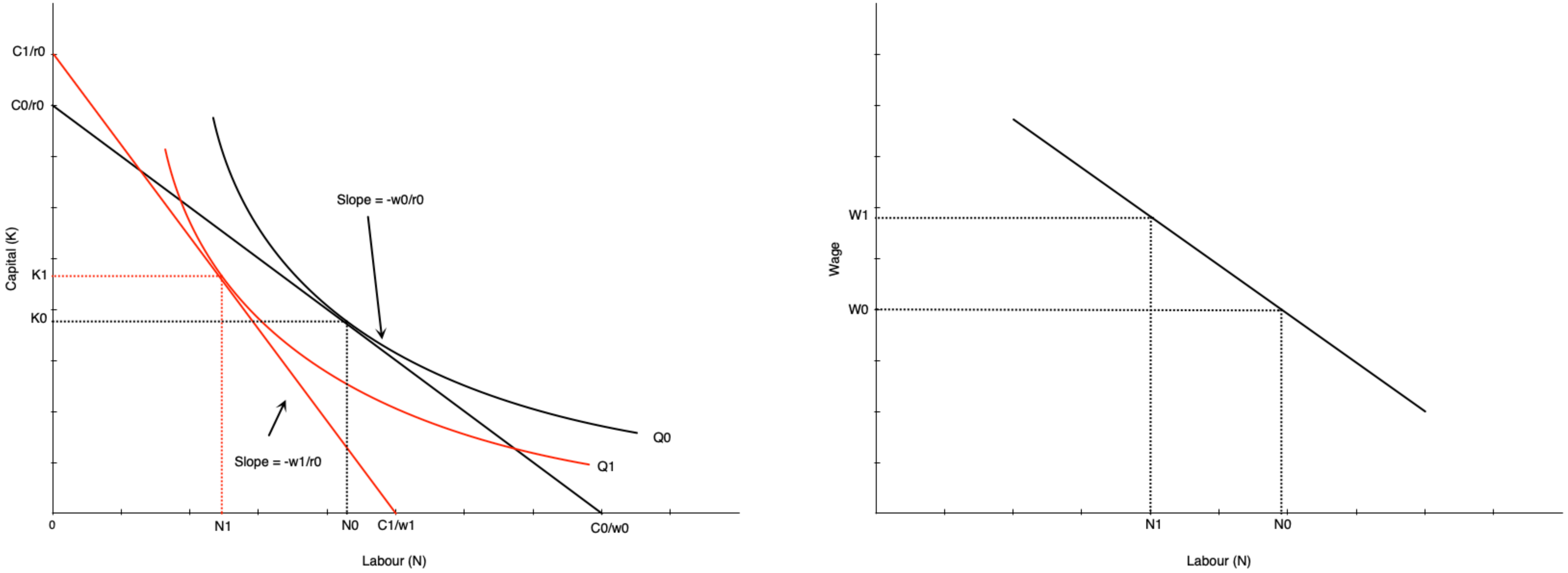


Effect of wage increase on labour:

- Wage goes up → isocost line steepens
- Firm scales down production to lower isoquant
- Tangency between new isoquant and new isocost shows new optimal labour and capital
- This involves less labour employed

Effect on capital: depends on substitutability between labour and capital (shape of isoquant)

Deriving Labour Demand



Deriving Labour Demand

The labour demand curve summarizes optimal choices of labour at different wage rates

- Previous slide shows two points on labour demand curve: (w_0, N_0) and (w_1, N_1)
- To derive full demand curve, vary wage rate to other levels

Ceteris Paribus

Demand curve shows amount of labour demanded at every wage all else equal — other determinants will shift the curve

Scale and Substitution Effects

Two Effects When Wage Changes

- Substitution effect: change in input mix due to change in relative input prices
- Scale effect: change in output due to change in cost of production

Substitution effect:

- Makes firms substitute capital for labour
- Leads to less labour demanded when wage rises (more capital demanded)

Scale effect:

- Comes from reduction in output
- As production becomes more expensive, firms produce less
- Leads to less labour demanded when wage rises (use less labour and capital)

Scale and Substitution Effects

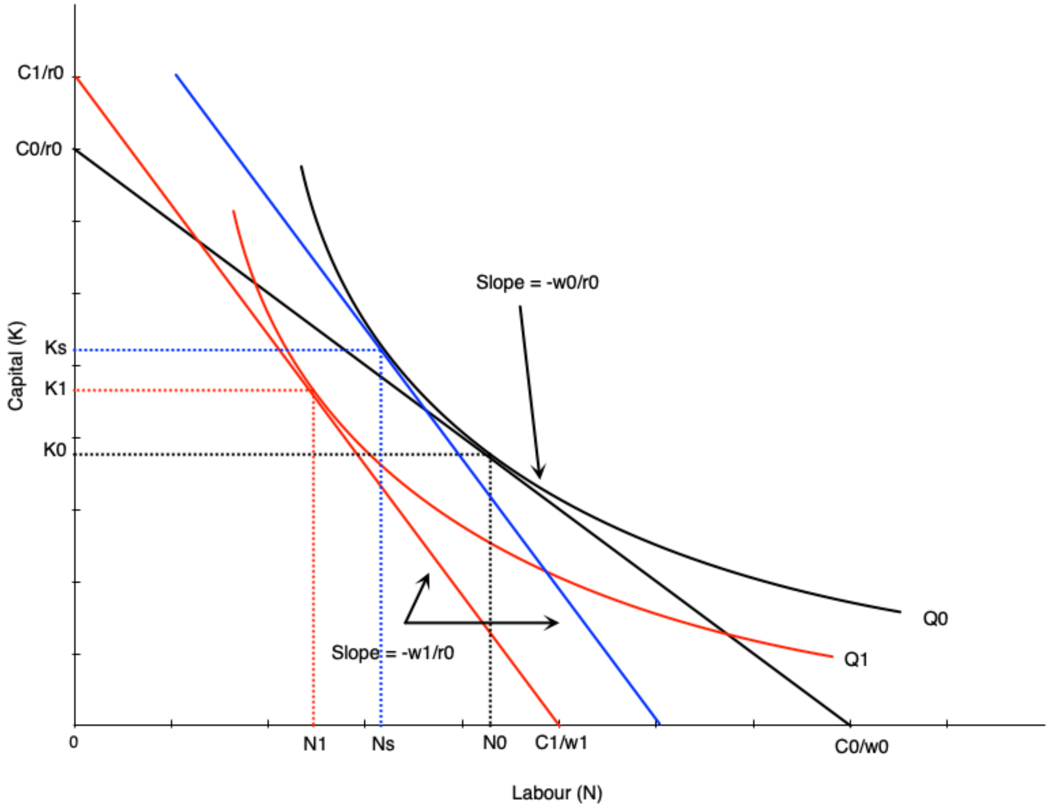
For Labour Demand

Scale and substitution effects work in the same direction — both reduce labour when wage rises

For Capital Demand

Scale and substitution effects work in opposite directions — effect is ambiguous

Scale and Substitution Effects



To find substitution effect:

- Find cost minimizing bundle at new wage and old output
- Blue line in graph
- Substitution effect: $N_0 \rightarrow N_s$

Scale effect:

- Change from optimum at new wage/old output to new wage/new output
- Shift from blue to red line in graph
- Scale effect: $N_s \rightarrow N_1$

Scale and Substitution Effects

Notice that scale and substitution effects have opposing effects on capital

When wage rises:

- Substitution effect → more capital used (firms substitute capital for labour)
- Scale effect → less capital used (firms reduce output, use less of all inputs)

⚠ **Warning**

Effect on capital is ambiguous

Relationship to Short-Run Demand

In the short run:

- Capital is fixed → no substitution effect

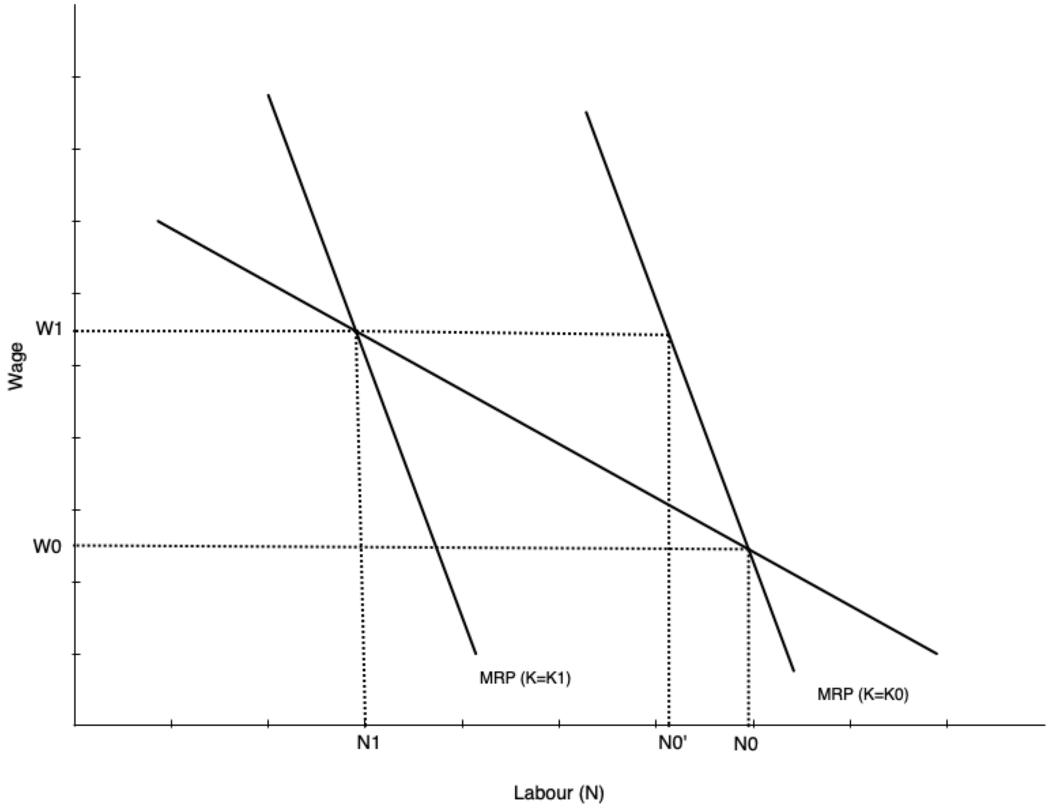
In the long run:

- Both substitution and scale effects occur

Key Result

Long-run labour demand is more elastic than short-run labour demand — firms can adjust all inputs in the long run, making them more responsive to wage changes

Relationship to Short-Run Demand



Graph shows short-run and long-run labour demand curves

- Start with capital fixed at K_0
- Wage goes from w_0 to w_1
- In short run: move along SR demand curve from N_0 to $N_{'}$
- In long run: firm substitutes capital for labour
- Short run curve shifts left
- Equilibrium at intersection of wage and new short run demand

Elasticity of Demand for Labour

Elasticity of Demand

Recall: elasticity of demand measures responsiveness of quantity demanded to changes in price

In this case, responsiveness of labour demanded to changes in wage rate

Elastic labour demand curve:

- Proportionately larger change in quantity than change in wage
- Curve is flatter

Inelastic labour demand curve:

- Proportionately smaller change in quantity than change in wage
- Curve is steeper

Factors Affecting Elasticity

Several factors affect elasticity of labour demand:

1. Ease of substitution between labour and other inputs
    - More/easier substitutes → more elastic demand
2. Elasticity of supply of other inputs
    - Inelastic supply of other inputs → harder to substitute → less elastic labour demand
    - Happens because other inputs become more expensive when firm tries to substitute
3. Elasticity of demand for the final product
    - Output responding more to price changes → more elastic labour demand
4. Labour share in total costs
    - Higher labour cost share → more elastic labour demand (operates via scale effect)

Evidence on Elasticity

In the USA:

- Elasticity between -0.15 and -0.75 (relatively inelastic)
- Elasticities have increased over time
- Long run elasticities are larger

In Canada:

- Elasticity between -0.1 and -0.6
- Depends on geography and industry

Key Result

Studies confirm that labour demand does slope down

Globalization and Offshoring

Introduction

Over time Canadian firms have faced increasing globalization

Has led to concerns over what happens to Canadian workers:

- People worry firms will move operations to countries with lower wages
- That this will destroy Canadian jobs
- Clear concern in the USA right now

Note

Effect of globalization on jobs is always a concern when trade agreements are reached — manufacturing is especially prominent, with automotive industry currently a focus

Effect of Trade on Single Market

Consider global competition with a specific product

Increased global competition lowers the price a product can sell for globally

In a competitive market, this reduces the value of the marginal product:

- Recall that $VMP = p \times MP_N$
- Shifts demand for Canadian labour inward

Effect of Trade on Single Market

Another approach: consider Canadian and foreign labour as two inputs to production

The optimal choice of Canadian and foreign labour occurs where:

$$\frac{MP_{N_C}}{MP_{N_F}} = \frac{w_C}{w_F}$$

The marginal product per dollar spent on foreign and Canadian workers are the same

Effect of Trade on Single Market

What effect does a fall in foreign wages have on Canadian labour demand?

Two effects occur:

- Substitution effect: firms substitute foreign workers for Canadian workers
  - Reduces demand for Canadian workers
- Scale effect: lower costs lead to more production
  - Increases demand for Canadian workers

Overall Effect is Ambiguous

- If close substitutes, substitution effect dominates → less Canadian labour demanded
- If scale effect dominates → could be more Canadian labour demanded

Effect of Trade on Single Market

> **Cost is Not the Only Factor**
>
> Productivity is also important — if Canadian workers are more productive, firms will still want to hire them

How productive are Canadian workers compared to foreign workers?

Next slides show international comparisons for OECD countries:

- Productivity, compensation, unit labour costs
- Changes over time

Effect of Trade on Single Market

TABLE 5.1 Productivity,[1] 2000, and Changes in Productivity, Hourly Compensation,[2] and Unit Labour Costs,[3] 2000–2018 (various OECD Countries)

Country	Productivity in 2000 (as % of Canada)	% Change in Productivity	% Change in Compensation	% Change in Unit Labour Costs
Korea	45	99	32	–23
Mexico	46	2	–44	
New Zealand	79	21	76	42
Japan	87	20	–38	–48
Australia	99	24	45	22
Canada	**100**	18	16	1
Spain	102	17	18	–1
United Kingdom	111	18	–5	–18
Finland	116	20	24	5
Italy	119	1	14	13
Sweden	121	28	17	–7
United States	123	30	2	–19
Germany	126	19	9	–6
France	129	18	19	6
Netherlands	135	15	19	3
Denmark	135	22	33	10
Switzerland	136	19	40	19
Belgium	143	14	21	5
Norway	164	16	46	27
Average	111	22	18	2

NOTES:

1. Output per hour. In descending order, from high to low, in terms of 2000 productivity
2. Defined as the sum of gross wages and salaries and employers' social security contributions. All figures are on an hours-worked basis and are exchange-rate-adjusted and therefore reflect exchange rates and compensation costs expressed in home country currency.
3. Hourly compensation per unit of output (i.e., adjusted for productivity).

SOURCE: Author's calculations based on Organisation for Economic Co-operation and Development, "OECD Productivity Statistics". Available online, accessed August 2020, https://www.oecd.org/sdd/productivity-stats/.

Effect of Trade on Single Market

FIGURE 5.8 Relative Trends in Labour Costs and Productivity: Canada Relative to the United States (base year 1989)

This figure shows relative trends in productivity and labour costs in Canada and the United States. Each point shows the ratio of a cost (or productivity) index in Canada to that in the United States, with 1989 as the base year. For example, in 2012 unit labour costs were 25 percent higher in Canada than in 1989, compared to the United States. Note that these figures do not compare the levels of costs between Canada and the United States, only their trends (relative to 1989).

NOTES: Points represent the relative value of indices of Canadian to U.S. labour costs and productivity (as defined in the notes in Table 5.1). The base year is 1989 (index = 100 for both Canada and the United States).

SOURCE: Author's calculations based on Organisation for Economic Co-operation and Development, "OECD Productivity Statistics". Available online, accessed August 2020, https://www.oecd.org/sdd/productivity-stats/.

Effect of Trade Across Sectors

> **Single Sector Misses Key Points**
>
> Trade openness affects the entire economy with adjustments across sectors

Recall from EC120: countries specialize in production of goods where they have comparative advantage

- Expands consumption possibilities

In a globalized market:

- Countries allocate more resources towards their comparative advantage sectors
- Labour moves from non-comparative advantage sectors to comparative advantage sectors

Evidence on Trade and Labour Demand

Many trade agreements with other countries

Most prominent in Canada: NAFTA and more recently CUSMA

These present opportunities to study effects of trade on labour demand

Evidence Shows

- About 10% of Canadian jobs lost in early 90s were due to free trade
- Productivity grew faster in manufacturing industries affected by trade

Non-Trade Demand Factors

At the same time trade is changing, the Canadian labour market is changing:

- Jobs have shifted away from manufacturing over long term
- Professional, managerial, and administrative service jobs have been increasing
- Flex or “gig” jobs have been increasing

All changing for various reasons:

- Technology
- Consumer preferences
- Demographics

Summary

Summary

Labour demand is a derived demand

- Firms hire labour to produce goods/services
- Optimal choice: $MRP_N = w$ (marginal revenue product equals wage)

Short run vs. long run:

- Short run: capital fixed, only scale effect
- Long run: both capital and labour variable, substitution and scale effects
- Long-run labour demand is more elastic

Labour demand slopes downward due to:

- Diminishing marginal product of labour
- Substitution and scale effects reinforce each other

Summary

Elasticity of labour demand depends on:

Ease of substitution, supply of other inputs, product demand elasticity, labour cost share

Globalization effects:

Ambiguous - substitution effect reduces Canadian labour demand, scale effect increases it